

UNIVERSITY OF ADEN
FACULTY OF ENGINEERING
TEST OF ODD SEMESTER FOR ACADEMIC YEAR (2020/2021)



Subject : Data Structure

DATE : 23/12/2020

Class : B2ITE / B2ITE

Student Name :

Reg.No:

Q1. Read and Tick (T) or (F)

- 1- Enqueue operation - circularly increase the REAR index by 1. (✓)
- 2- The **pop** operation modifies the stack - the next element becomes the top element. (✓)
- 3- We add elements to the queue, the Rear keeps on moving. (✓)
- 4- The following gives the value stored at the address pointed to by the pointer (a) is (a). (X)
- 5- If the operation **pop** is performed when the stack is full, it is said to be in **overflow** state. (X)
- 6- Linear data structure traverses the data elements sequentially. (✓)
- 7- The operator used to get value at address stored in a pointer variable is (&). (X)
- 8- Circular Queue is also known as ring buffer. (✓)
- 9- A stack is a data structure that uses the FIFO protocol. (X)
- 10- If **pop** is performed when the stack is empty - the stack is said to be in an underflow state. (✓)
- 11- If Queue is empty, then we cannot delete an element from Queue - "Underflow". (✓)
- 12- The following gives the memory address of integer variable a is &a. (✓)
- 13- The float data type used to represent a number without decimal point. (X)
- 14- In a circular queue, when the rear = n - 1 and a new element is to be added, the rear = 0. (✓)
- 15- Each queue has a data member, named as top, which points to the topmost element in the queue. (X)
- 16- The **getTop** operation modify the variable top. (X)

Q2. What output is displayed after the following segment of code executes?

```
stack <int> s;  
int a = 75, b = 2;  
s.push(8);  
s.push(a);  
s.push(a/b);  
b = s.top();  
s.pop();  
s.push(b);  
s.push(a+b);  
s.pop();  
while (!s.empty())  
{  
    cout << s.top() << endl;  
    s.pop();  
}
```

37 75 8

Infix notation	Prefix notation	Postfix notation
$A+B/C+(D * E)$	$++A/BC * DE$	$ABC/+DE * +$
$(A * B) + (C * D)$	$+ * AB * CD$	$AB * CD * +$
$A + (B * C)$	$+ A * BC$	$ABC * +$

Q4. What will be the output of the program ? (0xbeccfc7c, 0xbeccfc78, 0xbeccfc7c)

```
#include <iostream>
```

```
Using namespace std;
```

```
int main ( )
```

```
{
```

```
int x= 20;
```

```
int *p= &x;
```

```
*p=25;
```

```
*p=*p/x *50;
```

```
Cout<<&p<<endl; 08 — c78
```

```
Cout<<*p<<endl; 50
```

```
Cout<<x<<endl; 50
```

```
Cout<<p<<endl; — c7c
```

```
Cout<<&x<<endl; — c7c
```

```
return 0;
```

```
}
```

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Q1. Read and Tick (T) or (F)

- 1- A pointer - variable that stores address of an instruction. (X)
- 2- If value of FRONT variable is equal to SIZE - 1 then we can delete element from Queue. (✓)
- 3- After deleting topmost element from stack, we have to decrement the value of TOP by one. (✓)
- 4- Queue - the elements are added at one end and deleted from other end (LIFO). (X)
- 5- Enqueue operation - circularly increase the front index by 1. (X)
- 6- The **pop** operation modifies the stack - the newly inserted element becomes the top element. (X)
- 7- We add elements to the queue, the Rear keeps on moving. (✓)
- 8- The following gives the value stored at the address pointed to by the pointer (a) is (a). (X)
- 9- If the operation **pop** is performed when the stack is full, it is said to be in **overflow** state. (X)
- 10- Linear data structure traverses the data elements sequentially. (✓)
- 11- The operator used to get value at address stored in a pointer variable is (&). (X)
- 12- Circular Queue is also known as ring buffer. (✓)
- 13- If **pop** is performed when the stack is empty - the stack is said to be in an **underflow** state. (✓)
- 14- The following gives the memory address of integer variable a is &a. (✓)
- 15- In a circular queue, when the rear = n - 1 and a new element is to be added, the front = 0. (X)
- 16- stack is a data structure in which the elements are (popped or pushed) at only one end - LIFO. (✓)

Q2. What output is displayed after the following segment of code executes?

```

stack <int> s;
int a = 200, b = 10;
s.push(50);
s.push(a);
s.push(a*b);
b = s.top();
s.pop();
s.push(b);
s.push(a/b);
s.pop();
while (!s.empty())
{
    cout << s.top() << endl;
    s.pop();
}

```

2000 200 50

Q3. Convert :

Infix notation	Prefix notation	Postfix notation
A - B * C + (D / E)	+ - A * B C / D E	A B C * - D E / +
(A * B) + (C * D)	+ * A B * C D	A B * C D * +
A + B * C ^ D	+ A * B ^ C D	A B C ^ D * +

Q4. What will be the output of the program ? (0xbe88cc7c, 0xbe88cc78, 0xbe88cc7c)

```
#include <iostream>
```

```
Using namespace std;
```

```
int main ()
```

```
{
```

```
int x = 20;
```

```
int *p = &x;
```

```
*p = 25;
```

```
*p = *p * x;
```

```
Cout << &x << endl; 0x - cc7c
```

```
Cout << p << endl; 0x - cc7c
```

```
Cout << x << endl; - 625
```

```
Cout << *p << endl; 625
```

```
Cout << &p << endl; 0 - cc78
```

```
return 0;
```

```
}
```

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Q1. Read and Tick (T) or (F)

- 1- The following gives the value stored at the address pointed to by the pointer (a) is (a). (X)
- 2- Process of inserting an element in stack is called Push. (✓)
- 3- A queue follows - LIFO (✓)
- 4- If the operation **pop** is performed when the stack is full, it is said to be in **overflow** state (X)
- 5- Linear data structure traverses the data elements sequentially (✓)
- 6- The operator used to get value at address stored in a pointer variable is (&). (X)
- 7- Circular Queue is also known as Square Buffer. (X)
- 8- If Queue is empty, then we can delete an element from Queue - "Underflow". (X)
- 9- The following gives the memory address of integer variable a is &a. (✓)
- 10- The Integer data type used to represent a number without decimal point. (✓)
- 11- In a circular queue, when the rear = n - 1 and a new element is to be added, the rear = 0. (✓)
- 12- In stacks, we can access data elements from any intermediate positions. (X)
- 13- The **push** operation modifies the **top** - newly inserted element becomes the top. (✓)
- 14- Each queue has a data member, named as top, which points to the topmost element in the queue. (X)
- 15- The **pop** operation modifies the stack - the next element becomes the top element. (✓)
- 16- The **getTop** operation modify the variable top. (X)

Q2. What output is displayed after the following segment of code executes?

```

stack <int> s;
int a = 22, b = 44;
s.push(2);
s.push(a);
s.push(a + b);
b = s.top();
s.pop();
s.push(); (b);
s.push(a - b);
s.pop();
while (!s.empty())
{
    cout << s.top() << endl;
    s.pop();
}

```

66 22 2

Q3. Convert :

Infix notation	Prefix notation	Postfix notation
A+(B*C)	+A*BC	ABC*+
(A*B)+(C*D)	+*AB*CD	AB*CD*+
A*(B+C)/D	/*A+BCD	ABC+*D/

Q4. What will be the output of the program ? (0xbedc874, 0xbedc8c70, 0xbedc874)

```
#include<iostream>
```

```
Using namespace std;
```

```
int main ( )
```

```
{
```

```
int x= 100 ;
```

```
int *p= &x;
```

```
*p=100;
```

```
x=200;
```

```
*p=*p+x;
```

```
Cout<<&p<<endl; 0x — c70
```

```
Cout<<p<<endl; 0x — 874
```

```
Cout<<x<<endl; 400
```

```
Cout<<*p<<endl; 400
```

```
Cout<<&x<<endl; 0x — 874
```

```
return 0;
```

```
}
```

Good Luck ,,,

Q1. Read and Tick (T) or (F)

- 1- Dequeue operation circularly increase the rear index by 1. (X)
- 2- The following gives the value stored at the address pointed to by the pointer (a) is (a). (X)
- 3- The float data type used to represent a number with decimal point. (X)
- 4- If we can insert an element into stack, first we have to increment the value of variable TOP by one. (X)
- 5- If Queue is empty then we cannot delete an element from Queue - "overflow". (X)
- 6- After deleting topmost element from stack, we have to decrement the value of TOP by one. (X)
- 7- Each queue has a data member, named as top, which points to the topmost element in the queue. (X)
- 8- The *pop* operation modifies the stack - the next element becomes the top element. (X)
- 9- The *getTop* operation modify the variable top. (X)
- 10- A queue is a data structure that uses the Last In First Out protocol. (X)
- 11- If the operation *pop* is performed when the stack is full, it is said to be in *overflow* state. (X)
- 12- Non-linear data structure traverses the data elements sequentially. (X)
- 13- The operator used to get value at address stored in a pointer variable is (&). (X)
- 14- Circular Queue is also known as ring Buffer. (X)
- 15- The following gives the memory address of integer variable a is &a. (X)
- 16- The *push* operation modifies the *top* - newly inserted element becomes the top. (X)

Q2. What output is displayed after the following segment of code executes?

```
stack <int> s;  
int a = 33, b = 55;  
s.push(4);  
s.push(a);  
s.push(a + b);  
b = s.top();  
s.pop();  
s.push();  
s.push(a - b);  
s.push();  
while (!s.empty())  
{  
    cout << s.top() << endl;  
    s.pop();  
}
```

88 33 4

Q3. Convert :

Infix notation	Prefix notation	Postfix notation
$(A+B)*C$	$*+ABC$	$AB+C*$
$(A+B)*(C+D)$	$*+AB+CD$	$AB+CD+*$
$A*(B+C/D)$	$*A+B/C D$	$ABCD/+*$

Q4. What will be the output of the program ? (0xbe82dc74, 0xbe82dc70, 0xbe82dc74)

```
#include<iostream>
```

```
Using namespace std;
```

```
int main ( )
```

```
{
```

```
int x= 50 ;
```

```
int *p= &x;
```

```
*p=50;
```

```
x=500;
```

```
*p=*p+x;
```

```
Cout<<&x<<endl; 0x 74
```

```
Cout<<&p<<endl; c70
```

```
Cout<<x<<endl; 1000
```

```
Cout<<*p<<endl; 1000
```

```
Cout<<p<<endl; 0x c74
```

```
return 0;
```

Good Luck ,,,,